Replicating SwarmWorld with open-weight models

A 4-bit 27B reproduces the N=50 row, and serving numerics change the outcome

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Abstract

SwarmWorld (Pal, Wang and Buehler, 2026) reports that societies of language-model agents in a shared physical world build functional technologies and outperform the same number of agents searching alone, and attributes the gain to specific collective mechanisms. We replicate the paper's central comparison at N = 50, four conditions over the paper's four world seeds, using the authors' released engine, prompts, schema and commands unchanged, with open-weight models served on a lab fleet of 24 GB consumer cards. Four models were screened with the authors' validity-pilot protocol; a 27-billion-parameter dense model at 4-bit weight precision (Qwen3.8-27B, GPTQ) completed the row in about 26 hours on four RTX 3090s. Nine of the twelve seed-paired effects agree in sign with the paper's, including every portfolio-resilience gain and the discovery-frontier deficit of the shared-world conditions at this population size; the spatial self-organisation result reproduces in full; artifact counts are level with the paper's; and the authors' analysis pipeline, applied to our traces, draws the same picture. The 27B is the weaker solo agent (6 of 16 research milestones against the paper's 7 to 10) and the stronger society member (16 against 14), so the collective gap is wider, not narrower. Two findings arose on the way. The same 27B weights in two 4-bit builds differ by 0.17 in mean validity score and split 11-0 cell by cell, with the KV-cache precision that one checkpoint silently selects as the leading suspect, a larger difference than between the 27B and a 235-billion-parameter mixture-of-experts at 2-bit precision. And the released study traces reproduce every event, prompt and reported metric on the released engine, but not their hidden state digest, which points to code drift within one declared engine revision. Both are reported upstream. The replication package, Docker-only with per-checkpoint serving provenance, and the traces of every reported episode are released.

1 Introduction

SwarmWorld (Pal, Wang, and Buehler 2026) asks whether a society of language-model agents can invent and build functional technology in a shared physical world, and whether the collective mechanisms it provides (messages and public records, cross-agent inheritance of executable programs, and stigmergy through persistent artifacts) account for any advantage over the same agents searching alone. The paper answers both in the affirmative with a deterministic simulator, a strict action schema, a battery of engine-computed outcome metrics and a seed-paired design that compares each condition to an independent-search envelope on the same world. The authors released the engine, the analysis tools and the complete traces of every reported study (LAMM, MIT 2026a, 2026b), which makes the paper unusually replicable in principle: a second group can rerun the same commands and compare episode by episode.

In practice the paper was run through a hosted frontier model, and the question we set out to answer was a different one from "does the code run": does the result depend on that model, and can it be reproduced with open weights on hardware a small lab owns? The replication was governed by HAL8's AI Local Lab, whose fleet of 24 GB cards fixed the design space, and it was carried out in three steps. First, verify that

the released engine reproduces the released traces (it does, for everything observable, and does not, for one hidden digest; Section 4). Second, find an open-weight model that plays the game at all, by climbing a ladder of models and quantisations through the authors' own validity-pilot protocol (Section 5). Third, run the paper's N = 50 row with the model that passes and compare it seed by seed with the released episodes (Section 7).

This paper makes three contributions.

- **A replication of the N = 50 row with an open-weight 4-bit 27B.** Nine of twelve seed-paired effects agree in sign with the paper's, the spatial self-organisation result reproduces in full, and the collective gap is wider with the weaker solo agent. The row took about 26 hours on four consumer cards.
- **Two findings about reproducibility in this setting.** Serving numerics, specifically the precision of the KV cache that a checkpoint can select without the user's knowledge, moved the same model two rungs on the ladder, by more than the difference between a 27B and a 235B. And the released traces do not replay their hidden state digest on the released engine, although every event, prompt and metric does, which locates code drift inside a declared engine revision. Both were reported to the authors with evidence (Nedeljković 2026b, 2026a).
- **A Docker-only replication package with per-checkpoint serving provenance**, two opt-in configuration options for local servers that leave the paper's profiles byte-for-byte unchanged, a parallel runner for batching servers, and an acceptance test for OpenAI-compatible gateways, together with the traces of every episode reported here. The same model was also served through a metered marketplace gateway as a second path, to compare agents obtained through a marketplace with agents served directly.

Section 2 summarises SwarmWorld as far as the replication needs it. Section 3 describes the setup, Section 4 the engine verification, Sections 5 and 6 the models and the procedure. Section 7 gives the results, Section 8 discusses them, Section 9 lists the threats to validity, and Section 10 the open questions, recommendations and the work in progress. Costs are deliberately not reported, as the original paper reports none; throughput is given in tokens per second and hours.

2 SwarmWorld in brief

This section summarises the parts of SwarmWorld (Pal, Wang, and Buehler 2026) that the replication depends on. The original paper and the repository documentation (LAMM, MIT 2026a) remain the reference for everything else.

2.1 World, agents and the model interface

SwarmWorld couples language-model decisions to a deterministic physical simulator. The world is a grid of terrain cells (72 × 54 in the fifty-agent profile) with biomes, harvestable biological resources, processing stations and environmental fields for moisture, nutrients, solar exposure and contamination. Disturbances (contamination, drought, storm) strike the habitat at fixed intervals. The scientific task is habitat resilience: agents are to discover, fabricate, test and install material artifacts whose programs protect the habitat, and the simulator scores those artifacts by their measured field performance, not by what the agents say about them.

An agent does not call the model on every tick. On a fixed, staggered schedule (one *macroturn* per agent every 50 ticks in the fifty-agent profile, with phases spread across the population) the agent receives a local observation, its own retrieved private memory and whatever shared records its condition permits, and returns one JSON object validated against a strict schema: a private research-state update and a plan of up to twelve atomic actions. The plan enters a queue; one action per agent is attempted on every tick, and each action is checked against spatial, material, energetic and condition-specific constraints before it changes anything. Failed actions are recorded as rejections and become experience for the next macroturn. The schedule is treatment-invariant, so every condition grants exactly the same decision opportunities: a fifty-agent episode of 800 ticks contains 800 model calls, whatever the condition.

The model interface therefore consists of a prompt (whose hash the engine records in every trace header), the action schema (hashed the same way), a temperature of 0.7, an output budget of 4,096 tokens, a 60,000-character context budget and, for the paper's hosted model, a low reasoning effort. All outcome metrics are computed by the simulator from the recorded events; the model never sees a global reward.

2.2 Experimental conditions

Four conditions remove collective mechanisms without changing the task (Table 1). Independent search is the paper's control: N isolated one-agent worlds, reported as an endpoint-wise best-of-N envelope, so that a society is compared against the best of the same number of solo agents.

condition	shared world and artifacts	messages, public records, task claims, trading	cross-agent program forking and skill inheritance
full culture	yes	yes	yes
no communication	yes	no	yes
no explicit culture	yes	no	no
no program forking (validity-pilot protocol only)	yes	yes	no
independent search	no (N isolated worlds)	no	no

Table 1: The conditions and the mechanisms each retains, after Figure 3 of the original paper. "No program forking" is not among the paper's published N = 50 conditions; it is the fourth condition of the authors' validity-pilot protocol for local models (Section 6).

2.3 Endpoints

Each episode yields a small set of outcome metrics, all computed by the engine:

- **Discovery-frontier AUC**: the area under the curve of the best validated artifact performance over the episode.
- **Held-out resilience AUC**: at declared checkpoints the world is frozen, all agents are removed and eight exact clones are exposed to paired unseen disturbance schedules; the balanced service coverage integrated over the assay measures what the technology does without its inventors. The fifty-agent protocol uses eight held-out seeds (9201 to 9208).
- **Best artifact performance** and **portfolio resilience**: the best single artifact's field performance and the resilience of the whole installed portfolio at the end of the episode.
- **Artifacts**: the number of material systems built, and the collaborative fraction among them.
- **Research milestones**: sixteen ordered milestones from inspecting the environment through proposing a valid recipe, fabricating and testing microbatches, publishing evidence and pooling resources, to a causally composed artifact that passes field performance and shows emergent behaviour. The *research score* is the fraction of the sixteen reached. This is the metric our pilots are ranked on, because it is meaningful below the scale at which artifacts appear.

For seed-paired comparisons the paper reports each condition's effect as the condition's value minus the independent-search envelope on the same world seed. We use the same convention and the same four seeds (3201 to 3204).

2.4 Protocols

The paper's main comparison at N = 50 is two studies run from one command each: *full culture and independent search* (study 1) and the *culture ablations*, no communication and no explicit culture (study 2), each over four seeds and 800 ticks. The repository additionally documents a *validity pilot* for local models,

four conditions at N = 4 and N = 16 over two development seeds and 400 ticks, and we added a 32-call *quick pilot* as a format check. Table 2 lists them; Section 6 says what each can show.

	quick pilot	validity pilot	N = 50 row	full paper matrix
conditions	full culture	full, no communication, no program forking, independent search	full and independent search (study 1); no communication and no explicit culture (study 2)	as N = 50, plus N = 100 and N = 200 studies and a 3,200-tick crossover
population	4	4 and 16	50	50, 100, 200
world seeds	3001	3001, 3002	3201 to 3204	3201 to 3204
held-out seeds	9001, 9002	9001 to 9004	9201 to 9208	9201 to 9208
ticks	400	400	800	800 (3,200 for the crossover)
cells (episodes)	1	16	8 per study, 16 for the row	89 episodes plus member traces
model calls per cell	32	32 at N = 4, 128 at N = 16	800	800 to 6,400
model calls in total	32	1,280	12,800	166,465
prompt size (tokens)	about 4.5k	4k to 7k	about 11k	up to about 13k

Table 2: The three protocols we ran and the paper's complete matrix. Model calls per cell follow from the fixed schedule: agents × ticks / 50. Development seeds (3001, 3002) are never pooled with the paper's seeds.

3 Replication setup

3.1 Hardware and hosts

The replication ran on HAL8's local lab fleet (Table 3). The simulator itself is CPU-only and ran on one host throughout; model inference ran on whichever cards the model under test needed.

host	GPU	role in this work
stribor-01	1 × NVIDIA RTX A5000, 24 GB	simulation driver (dual Xeon, 256 GB RAM); vLLM for the single-card pilots; Step 0
zmaj	1 × NVIDIA RTX A5000, 24 GB	second vLLM replica (pool tests)
vuk	1 × NVIDIA RTX 3090, 24 GB	llama.cpp master for the 235B mesh; vLLM replica for the N = 50 row
svarog-01, -02, -03	1 × NVIDIA RTX 3090, 24 GB each	llama.cpp RPC workers for the 235B mesh; vLLM replicas for the N = 50 row

Table 3: Hosts and cards. All cards are 24 GB Ampere parts, which fixes the design space: a 27B model fits one card only at 4-bit weights, and the 235B mixture-of-experts fits four cards only at 2-bit weights.

3.2 The package: engine, patches and provenance

Everything runs from one Docker Compose package, with no Python environment on any host. The simulation image is `python:3.12-slim` with the released SwarmWorld source vendored at commit 6af7ae9 (release provenance: source commit 504b7186 of 2026-08-29, engine revision 9) and installed with the authors' own extras. Model servers are the official `vllm/vllm-openai` image (vLLM 0.29.0) and llama.cpp's `llama-server`;

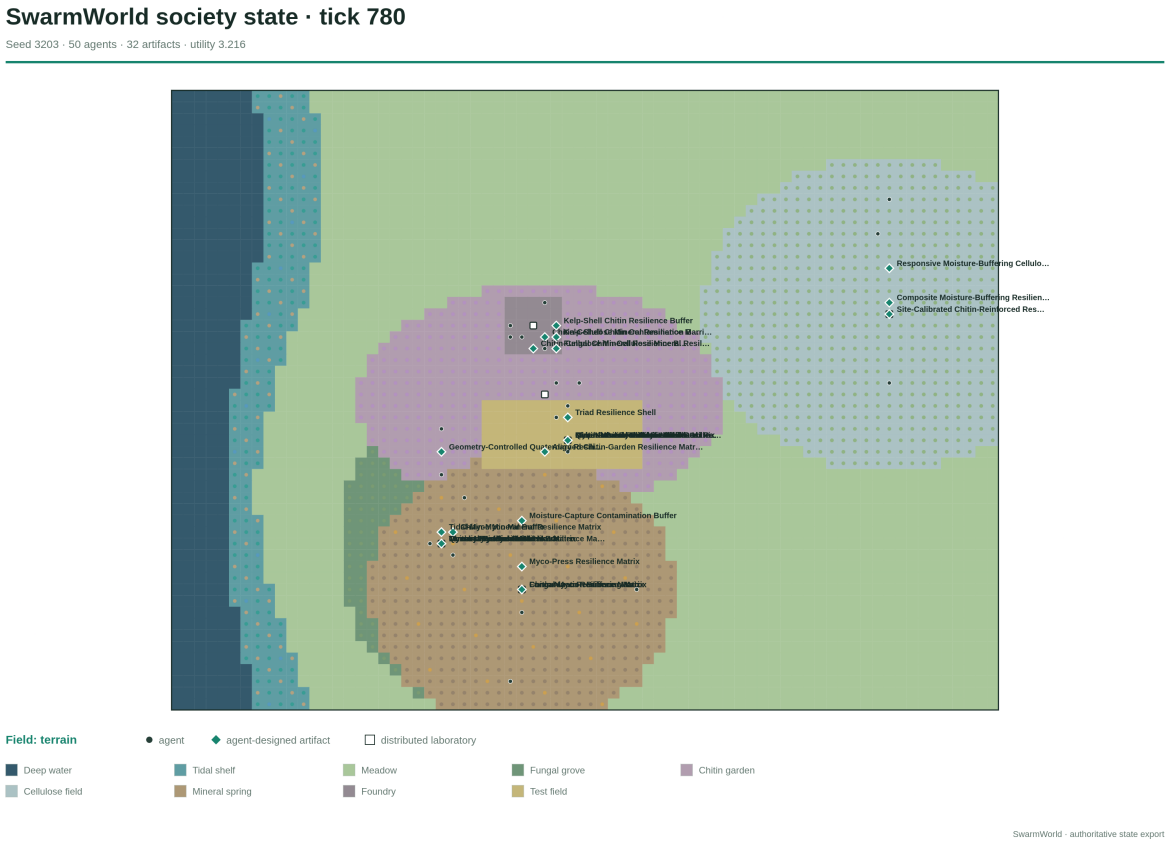


Figure 1: The authors' observatory playing back our full-culture episode of seed 3203 (Qwen3.8-27B GPTQ Int4) at tick 780: the fifty agents (dots), the 32 artifacts standing at that tick (diamonds, 35 by tick 800) and the distributed laboratory stations (squares) on the terrain map. Exported from the viewer's paper-export control; artifact labels overlap where the society has clustered its technologies.

an nginx container balances several vLLM replicas; the authors' browser observatory runs from a Node container. The source tree is bind-mounted, so the two patches below are live without a rebuild and the pristine files are kept beside them.

The engine, prompts and action schema are unmodified. Two options were added to the configuration, both off by default and both absent from the paper's profiles, so that the paper's own runs would be byte-for-byte the original:

- `llm.strict\_id\_patterns`: adds a regular-expression `pattern` to the three free-string ID fields of the action schema (`reply\_to`, `target\_agent\_id`, `target\_artifact\_id`), so that constrained decoding cannot emit the literal placeholder "NONE" that small models copy from the prompt into those fields. The paper's own no-reply condition pins `reply\_to` the same way. With the flag on, the schema hash recorded in every trace header changes, so traces made with it are self-labelled.
- `llm.transport:\chat\_completions` with `llm.extra\_body`: posts the same prompt and the same schema object to the server's `/chat/completions` endpoint as `response\_format` instead of to `/responses` as `text.format`. llama.cpp's Responses endpoint silently drops the schema; through the chat endpoint it is enforced. Usage counts are mapped back to the Responses names so that traces and the compute audit stay uniform. `extra\_body` carries the switch that turns a thinking model's reasoning off.

A third change is diagnostic only: when a reply is cut off at the output budget, the error recorded in the trace now carries the length, head and tail of the partial reply, so that runaways can be classified afterwards. All three were submitted upstream as a pull request (Nedeljković 2026a); the unit tests added with it verify that the schema hash with the flags off equals the unpatched module's.

Every run writes a manifest beside its traces with the package version, the prompt, schema and configuration hashes, the served model name, the server version, the transport and the sampling settings, and every run and pilot writes into a new time-stamped folder, so earlier attempts remain on disk as a trail. The engine hashes the prompt and the schema per condition, since the conditions remove fields and sections; the prompt hash of every local trace equals the released trace's for the same condition (`0bd15c0f…` for full culture, `b384028f…` for no communication and independent search, `b88c62e7…` for no explicit culture), and the action-schema hash differs from the released value only through the ID-pattern flag. Rendered configurations are named by the hash of their content, and a per-checkpoint registry records what the traces cannot: weights repository and revision, quantisation recipe, kernels, context length, KV-cache size and precision, server flags and measured throughput (Appendix A).

3.3 Running at scale: parallel runner and balancer

The authors' `research-study` command runs the cells of a study one after another. For a batching server that is a waste, so the package runs each cell (one condition, one population size, one seed) as its own `research-study` process with the authors' exact arguments, and merges the per-cell summaries into the study summary the authors' analysis tools expect, refusing to merge if the cells' manifests differ in anything but the cell itself. A heartbeat reads the partially written traces and reports ticks, calls, errors, seconds per call and the estimated finish. Finished cells are kept across restarts and only cells in flight are redone. On the 24 GB cards the number of streams is set by the KV cache: with the fifty-agent prompts of 6k to 15k tokens, two concurrent requests per card use half to two thirds of the 37.5k-token bf16 cache of the 27B build, and a third stream queues; four cards therefore serve eight streams, and the eight cells of an $N = 50$ study run at once. A least-connections nginx balancer in front of the replicas spreads the streams evenly; the simulator sees one endpoint.

3.4 Two serving paths: direct and through the Vector marketplace

Every run reported in Section 7 took one of two paths to the same model servers. The *direct* path points the simulator's `base\_url` at the balancer, which fronts the lab's vLLM replicas. This is how the ladder and the $N = 50$ row were produced.

The second path goes through the Vector marketplace, an OpenAI-compatible gateway (`https://api.ma`

rketplace.vector.apexfusion.org/openai/v1) that fronts third-party model suppliers, meters every request by tokens and settles usage on chain. A supplier registers an OpenAI-compatible endpoint under an alias and receives traffic the gateway forwards; a client uses the gateway exactly as it would use the OpenAI API, with a marketplace key in place of the provider key. We registered the lab's own balancer as a chat-completions supplier under the alias `swarm-qwen38-27b-gptq4`, so that the same weights, server and settings answer on both paths and the only difference is the gateway in between. On the simulator's side the switch is one profile whose `base\_url` is the gateway and whose `api\_key\_env` names the marketplace key; because the gateway forwards only standard OpenAI fields, the reasoning switch that the direct profile sends in `extra\_body` is instead applied server-side through a chat template with thinking disabled, on all four replicas. A five-probe acceptance test in the package (plain chat; `response\_format` of type `json\_object`; strict `json\_schema`; a schema the model would not satisfy by itself, to prove enforcement rather than tolerance; and a Responses-style request with `text.format`, to prove translation) passes against the balancer directly and through the gateway, which establishes that the schema the simulation depends on reaches the supplier intact. The purpose of the second path is to compare the results obtained from agents served through the marketplace with those obtained directly; Section 7.6 reports how far that comparison has been taken.

4 Engine verification (Step 0)

Before any model call, the package verifies that the released engine, as built in our image, behaves as the authors' documentation says it does. The step runs in about five minutes and makes no model calls.

4.1 Tests and scripted replay

The authors' test suite passes in the image (181 tests with the analysis extras, 165 in the engine-only image), and their `doctor` and lint checks pass. A scripted-policy episode recorded in the image replays bit-exactly on the same image: every stored authoritative-state digest verifies. This establishes that our own traces are internally reproducible, which Section 7 depends on.

The released dataset (LAMM, MIT 2026b) is downloaded and every file is verified against the hashes in its manifest (one documentation file differs from its recorded hash; the data files do not).

4.2 Replaying the released traces

The authors' reproducibility note states that a released trace replays deterministically on the released engine, and the README uses the fifty-agent full-culture trace of seed 3201 as its example. On our image that command fails at snapshot 32 (tick 744 of 800) with a deterministic-state mismatch, after snapshots 1 to 31 verify. The finding generalises (Table 4): six of the seven released traces we tried diverge between tick 648 and tick 744, and one replays to the end.

What matters for a replication is which parts of the trace the engine does reproduce, and the answer is everything observable. With a single-pass replay that applies the recorded model traces and macroturn grants exactly as the replay command does, we checked the seed 3201 trace in full: the rendered-snapshot digests match for all 35 snapshots, including the four after the state divergence; the events the engine emits equal the recorded events on all 800 ticks; the agent prompts rebuilt from the replayed state equal the recorded prompts for all 791 macroturns whose prompt was not truncated (the retrieved memory set is identical, and in 128 macroturns its order differs because the retrieval statistics are live-policy state that replay does not rebuild); and the per-episode summary rebuilt from the final state equals the released `study-summary.json` on every field except the retrieval statistics and the cross-agent program-fork count, whose definition the released code upgraded after the studies were run (the released analysis module contains the upgrade path).

Hash randomisation, four numpy versions and floating-point drift were ruled out; a perturbation of one field cell at tick 1 changes the digest at the very next snapshot, so the digest is exact and a platform difference would surface early, not at tick 700. The divergence is consistent with a change, between the studies' commits and the public release, to a bookkeeping field that enters the hidden authoritative state but not the rendered snapshots, events, observations or metrics. We reported it upstream with the per-trace evidence (Nedeljković

trace	source commit in header	state digest verifies through	diverges at
N = 50 full, seed 3201	83b9e7ac	tick 720	tick 744
N = 50 full, seed 3202	83b9e7ac	tick 696	tick 720
N = 50 full, seed 3203	83b9e7ac	tick 624	tick 648
N = 50 no explicit culture, seed 3204	83b9e7ac	tick 648	tick 672
N = 100 no communication, seed 3204	76fa2044 (dirty)	tick 624	tick 648
N = 200 no explicit culture, seed 3202	738cef68	tick 624	tick 648
N = 100 full, seed 3201	ab68bab8	tick 800 (all 35 snapshots)	never

Table 4: Replay of released traces on the released engine (commit 6af7ae9). All nine released study manifests record engine revision 9, but six different source commits, none of them in the public history.

2026b); the consequence for this paper is that comparisons use the released per-episode metrics, which the released engine reproduces, and treat the program-fork fraction as version-sensitive.

5 Models and serving

5.1 The model ladder

We did not pick one local model and run the paper with it. The question was which open-weight model, if any, plays this game at all, so the models were climbed as a ladder, each rung admitted to the next protocol only if it cleared the one before (Table 5). The paper's hosted model, gpt-5.6-luna, enters only through its released traces; we have not rerun it.

Gemma 4 E4B is the model the authors ship a local profile for; it is the floor of the ladder. Qwen3-235B-A22B was the largest open-weight model the fleet can hold, and the first that played the game. Qwen3.8-27B is the model that completed the row; it is a dense hybrid of gated-delta-network linear-attention layers and full-attention layers, with a vision encoder that we disabled to fit the 24k context in 24 GB. Three 4-bit builds of it were tried; two ran.

5.2 Serving details that matter

Several serving facts changed either feasibility or results and are therefore part of the method.

- **Structured output.** vLLM's default grammar backend (xgrammar) rejects the `pattern` keyword our ID constraint uses; the guidance backend accepts it, and its option to forbid free whitespace stopped the whitespace runaways that Gemma produced under the grammar. All vLLM runs use guidance.
- **Thinking.** The Qwen models are thinking models. With thinking on, the reasoning consumed the output budget and the plan arrived truncated (18 of 33 plans invalid in the first 235B attempt). All Qwen runs have thinking off, either by a request field or, on the marketplace path, by the server's chat template.
- **Sampling.** The two 27B checkpoints ship a generation configuration with top-k 20 and top-p 0.95, which vLLM applies under the request's temperature of 0.7. The paper's hosted calls carry no top-k. We kept the checkpoint defaults and record them here as a known difference.
- **KV cache precision.** The INT4 checkpoint ships a KV-cache quantisation scheme (8-bit float, static per-tensor scales), and vLLM honours it: the cache for that build is stored as fp8 and FlashInfer is selected. The GPTQ checkpoint has no such scheme, so its cache is bf16 and FlashAttention 2 is selected. Neither was a choice we made; both follow from the checkpoint files, and we noticed the difference only

model	weights	active parameters	server	cards	context	KV cache	transport
gpt-5.6-luna (paper)	hosted	undisclosed	OpenAI Responses API	hosted	60,000-character budget	n/a	Responses, strict schema, reasoning effort low
Gemma 4 E4B	bf16, `google/gemma-4-E4B-it`	about 4B effective	vLLM 0.29.0	1 × A5000	32,768	bf16, 5.6 GB (270k tokens)	Responses, guidance grammar
Qwen3-235B-A22B	GGUF UD-Q2_K_XL (Unsloth dynamic 2-bit)	22B of 235B	llama.cpp `llama-server`, RPC pipeline over four nodes	4 × RTX 3090	32,768	q4_0 keys and values	chat completions, thinking off
Qwen3.8-27B, INT4	compressed-tensors w4a16, `RedHatAI/Qwen3.8-27B-INT4`	27B (dense, hybrid)	vLLM 0.29.0, FlashInfer	1 × A5000	24,576	**fp8 e4m3** from the checkpoint's KV scheme, 2.85 GB (65.9k tokens)	chat completions, thinking off
Qwen3.8-27B, GPTQ	GPTQ Int4, `Max73333/Qwen3.8-27B-GPTQ-Int4-V100`	27B	vLLM 0.29.0, FlashAttention 2	1 × A5000; 4 × RTX 3090 for the row	24,576	**bf16**, 2.82 GB (37.5k tokens)	chat completions, thinking off
Qwen3.8-27B, AWQ	AWQ 4-bit, `barrydeen/Qwen3.8-27B-AWQ-4bit`	27B	vLLM 0.29.0	1 × A5000	—	—	never loaded

Table 5: The checkpoints and how each was served. Both 27B builds load to 16.8 GiB on a 24 GB card with the vision encoder disabled; the AWQ build does not load at all because vLLM 0.29 instantiates most of its layers in bf16 (its quantisation configuration does not match the model's hybrid linear-attention module names), which needs 54 GB. Appendix A gives the complete registry.

after the two validity pilots had split (Section 7.4). Forcing fp8 storage on the GPTQ build would use unit scales and change its numerics, and Ampere cards have no hardware fp8 arithmetic, so the fp8 cache is storage precision only.

- **Mixture-of-experts over RPC.** On the four-card llama.cpp mesh two parallel slots gave 7.35 generated tokens per second in aggregate against 22 for a single stream, because a second sequence reads a nearly disjoint expert set per token and doubles the weight traffic per step. The 235B therefore ran one episode at a time, which is why it stops at the validity pilot.
- **Concurrency on one card.** vLLM batching gave 2.3 × the aggregate throughput at four streams on the A5000 for the INT4 build, whose fp8 cache holds 2.7 concurrent 24k-token requests. The GPTQ build's bf16 cache holds 1.5, and four streams on it queued inside the server until the client's 300 s timeout expired (29 retried calls in its validity pilot). The profile's timeout was raised to 900 s and the N = 50 row ran at two streams per card.

5.3 Adaptations for local models

Three behaviours of small models required the options described in Section 3, and each is visible in the traces as a format or transport fact rather than a change of the task:

1. **Placeholder identifiers.** The prompt says to use "NONE" when a resource or artifact name is irrelevant. Gemma 4 E4B copied the literal into the message-reply and target-artifact fields in 29 and 22 of 42 actions, and the engine rejects any action whose reply target is not a delivered message, so 27 of the first pilot's 32 rejections and most of the society's inertia came from one formatting habit. The schema patterns remove it at decoding time; the paper's model leaves the fields empty on its own.
2. **Dropped schema on the Responses endpoint.** llama.cpp's `/v1/responses` converts to chat completions internally and discards the schema, so the JSON came back unconstrained. The chat-completions transport sends the same schema object where that server enforces it.
3. **Reasoning in the output budget.** Thinking off, as above.

A fourth adaptation is to the serving fleet rather than the model: the paper's own concurrency setting (`llm.concurrency`) parallelises calls within one episode, which a fixed-schedule society of 50 agents rarely exercises, so throughput comes from running cells in parallel across cards (Section 3).

6 Procedure

Every model went through the same sequence, and each stage reused the authors' `research-study` command with their arguments; only the model profile and the seeds and ticks named below differ.

Quick pilot (full culture, N = 4, seed 3001, 400 ticks, 32 calls). A format check that takes minutes: does the model produce valid plans, how long are they, how many actions does the engine reject and why, how fast is a call. It decides nothing about the model's ability; it decides whether the serving stack is right. Three quick pilots were needed before Gemma 4 E4B ran clean and two before the 235B did, and each fix is one of the options of Section 3.

Validity pilot (the authors' local-model protocol: full culture, no communication, no program forking and independent search; N = 4 and N = 16; development seeds 3001 and 3002; 400 ticks; 1,280 calls; the Gemma pilot, run before the parallel runner existed, had no explicit culture as its fourth condition instead). This is the smallest protocol on which the paper's central contrast, a society against independent search, can be observed, and it is also the protocol on which two builds of the same weights can be compared cell by cell, since the sixteen cells share conditions, sizes, seeds and prompts. With two seeds it cannot establish a direction for an effect; it ranks models and screens them for the row. The sixteen cells ran in parallel, four streams on one card for the 27B builds and one at a time on the mesh for the 235B. Development seeds are never pooled with the paper's seeds.

The N = 50 row (study 1: full culture and independent search; study 2: no communication and no explicit culture; seeds 3201 to 3204; 800 ticks; held-out seeds 9201 to 9208; 12,800 calls). The paper's main comparison, run with the paper's two study commands, so that every episode has a released counterpart with

the same seed and the seed-paired effects can be read off directly. The eight cells of each study ran at once as eight streams over the four RTX 3090s behind the balancer, two streams per card. Independent search at N = 50 consists of fifty isolated one-agent episodes per seed, recorded as member traces and reduced by the authors' analysis to the best-of-N envelope.

Three further rules held throughout. First, nothing is retried by hand: the engine's own retry policy handles transient provider errors and freezes world time during an outage, exactly as in the paper, and every error is recorded in the trace with its kind. Second, every analysis number in Section 7 comes from the authors' tools (`analyze-study`, `analyze-mobility`, `trace-report`, `technology-dossiers`) applied to our traces, or from two small scripts in the package that tabulate the study summaries and join them to the released `episodes.csv` by condition, population and seed. Third, all runs that are compared with each other ran on the same package version, the same engine revision and, per condition, the same prompt and schema hashes, and the manifests prove it; where a manifest field drifted between the cells of one run (a timeout changed by a stale configuration during the 235B validity pilot), the drift is recorded in the merged summary and discussed in Section 9.

Table 6 gives the scale of each run that enters the results.

run	model	protocol	calls	errors	streams	wall time
E4B validity	Gemma 4 E4B	validity pilot	1,280	0	1	2 h 18 min
235B quick	Qwen3-235B-A22B Q2	quick pilot	32	0	1	26 min
235B validity	Qwen3-235B-A22B Q2	validity pilot	1,282	11	1	25 h 10 min
27B INT4 quick	Qwen3.8-27B INT4	quick pilot	32	0	1	14 min
27B INT4 validity	Qwen3.8-27B INT4	validity pilot	1,280	1	4	3 h 52 min
27B GPTQ quick	Qwen3.8-27B GPTQ	quick pilot	32	0	1	18 min
27B GPTQ validity	Qwen3.8-27B GPTQ	validity pilot	1,309	29	4	7 h 15 min
27B GPTQ N = 50, study 1	Qwen3.8-27B GPTQ	full culture and independent search	6,402	35	8	16 h 32 min
27B GPTQ N = 50, study 2	Qwen3.8-27B GPTQ	no communication and no explicit culture	6,400	1	8	9 h 20 min
27B GPTQ, marketplace quick	Qwen3.8-27B GPTQ via the marketplace	quick pilot	32	0	1	32 min

Table 6: The runs behind Section 7. Errors are model calls the engine recorded as failed and retried; their kinds are classified in Section 7.5. Call counts above the nominal figure are retries. Wall time is the whole run including the held-out assays.

7 Results

Four local models entered the ladder; one completed the paper's N = 50 row. We report the ladder first (Section 7.1), then the row against the released episodes (7.2), the authors' analysis pipeline applied to our traces (7.3), the two-build comparison that produced the serving finding (7.4), throughput and errors (7.5), and the marketplace path (7.6). Every number is computed by the engine or the authors' tools from traces in the package; Appendix B lists every cell.

7.1 The local-model ladder

Figure 2 and Table 7 give the validity pilots. The four models order cleanly on the research score, and the ordering is not the ordering by size.

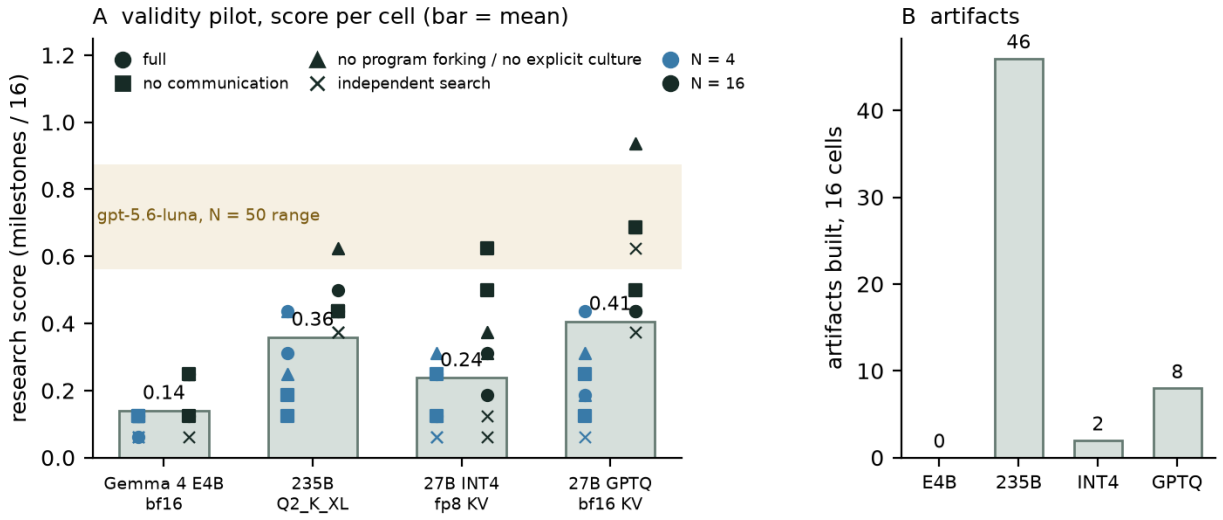


Figure 2: Validity pilots, sixteen cells per model. (A) Research score per cell (markers) and the mean over cells (bar); blue markers are N = 4, black N = 16. The band is the range of the paper's gpt-5.6-luna episodes at N = 50, a different protocol, shown for scale only. (B) Artifacts built over the sixteen cells.

model	mean score	score range	milestones	artifacts	cells with artifacts	actions per call	rejected actions
Gemma 4 E4B bf16	0.14	0.06 to 0.25	1 to 4	0	0	1.3	15.9 %
Qwen3-235B-A22B Q2_K_XL	0.36	0.125 to 0.625	2 to 10	46	14	5.8	18.4 %
Qwen3.8-27B INT4 w4a16 (fp8 KV)	0.24	0.06 to 0.625	1 to 10	2	2	7.2	3.0 %
Qwen3.8-27B GPTQ Int4 (bf16 KV)	0.41	0.06 to 0.94	1 to 15	8	7	8.9	6.5 %

Table 7: The validity pilots. Score is milestones reached over sixteen; actions per call counts the non-waiting actions the engine executed per model call (the paper's model executes about 5.5); rejected actions are planned actions the engine refused, a format and game-sense measure rather than an outcome.

Gemma 4 E4B is the floor: one-action plans, agents waiting on 97 % of ticks, no artifact in any cell, and conditions that do not separate because there is nothing cultural to lose. The 235B is the builder: artifacts in 14 of 16 cells, up to eight in one, with the highest rejection rate of any model because it attempts operations the world refuses (operating away from a station, testing before a batch exists). The 27B GPTQ build climbs furthest: it reaches the most milestones and one of its cells, no program forking at N = 16, seed 3001, reaches

15 of 16, above the range of the paper's N = 50 episodes. Compared cell by cell with the 235B it is higher in 6 cells, equal in 2 and lower in 8, so the two have different profiles rather than a rank: the 235B builds more and reaches less. The paper's central contrast appears at this scale only weakly: with the 235B, full culture is above independent search in both N = 16 seeds by a small margin; with the 27B GPTQ, independent search at N = 16 is level with full culture. Two seeds cannot establish a direction either way, which is what the row is for.

7.2 The N = 50 row against the paper

Qwen3.8-27B GPTQ ran the paper's two study commands over the paper's four seeds: 16 episodes, 12,802 model calls, 36 recorded errors. Each episode has a released counterpart with the same world seed, so the comparison is seed-paired throughout (Figure 3, Table 8).

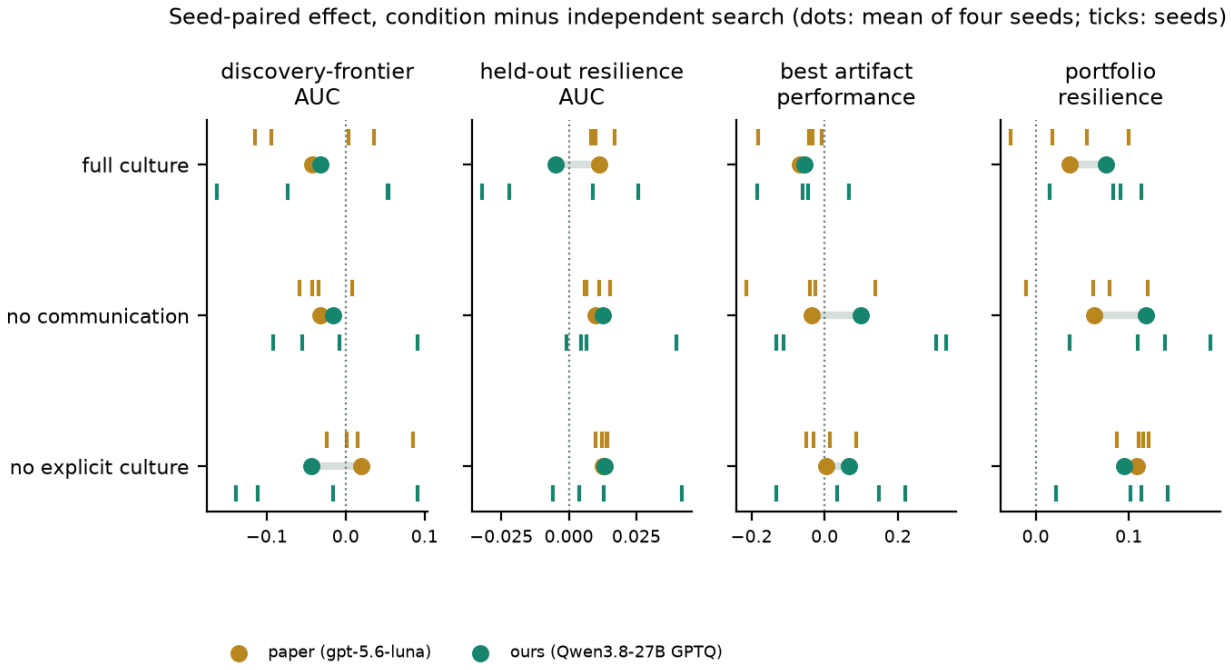


Figure 3: Seed-paired effects at N = 50, condition minus the independent-search envelope on the same seed, ours (green) against the paper's gpt-5.6-luna episodes (gold). Dots are the mean over the four seeds, ticks the individual seeds.

condition	discovery-frontier AUC	held-out resilience AUC	best artifact performance	portfolio resilience
full culture	−0.032 | −0.042	−0.005 | +0.011	−0.056 | −0.067	+0.076 | +0.037
no communication	−0.016 | −0.032	+0.012 | +0.010	+0.099 | −0.035	+0.118 | +0.063
no explicit culture	−0.044 | +0.019	+0.013 | +0.013	+0.067 | +0.005	+0.095 | +0.108

Table 8: Seed-paired effects, mean over seeds 3201 to 3204, ours | paper. Nine of the twelve agree in sign. The three that do not (full-culture held-out resilience, no-communication best artifact performance, no-explicit-culture discovery frontier) are within a few hundredths of zero on at least one side.

The paper's clearest collective gain at N = 50 is portfolio resilience, positive for every culture condition; it is positive for every culture condition here too, and larger. The paper's negative discovery-frontier effect of full culture and no communication at N = 50 (the conditions trail the independent envelope on this endpoint at this population size, before overtaking it at N = 100) reproduces in sign and magnitude. Best artifact

performance under full culture is below the envelope on both sides by the same amount. The disagreements are small effects: the paper's +0.011 and +0.019 become −0.005 and −0.044 here, and its −0.035 becomes +0.099.

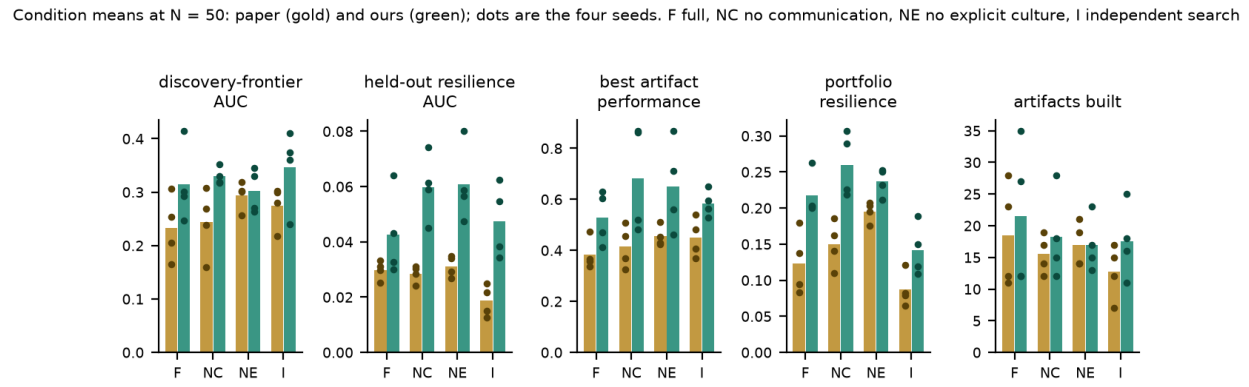


Figure 4: Condition means at N = 50, paper (gold) and ours (green); dots are the four seeds.

The condition means (Figure 4, Table 9) show where the two models differ. Every outcome metric is higher with the 27B than with the paper's model in every condition, including independent search: best artifact performance 0.53 to 0.68 against 0.38 to 0.45; discovery frontier 0.30 to 0.35 against 0.23 to 0.29; held-out resilience about twice the paper's. Artifact counts are level: full culture 21.5 against 18.5, no communication 18.3 against 15.5, no explicit culture 17.0 against 17.0, and the per-seed full-culture counts, 12, 27, 35 and 12, sit beside the paper's 12, 28, 23 and 11. Since the effects are differences between conditions on the same seed, a uniformly higher baseline does not by itself move them, and that is what Table 8 shows.

condition	milestones of 16 (ours)	milestones (paper)	artifacts (ours)	artifacts (paper)	collaborative fraction (ours)	collaborative fraction (paper)	mean distance travelled (ours)	(paper)
full culture	16, 16, 16, 16	14, 14, 14, 14	12, 27, 35, 12	12, 28, 23, 11	0.99	0.67	52.7	34.9
no communication	12, 12, 11, 12	—	12, 18, 15, 28	14, 17, 19, 12	0.57	0.26	62.0	35.1
no explicit culture	12, 12, 12, 12	—	13, 17, 23, 15	14, 21, 19, 14	0.61	0.22	60.4	36.2
independent search	6, 6, 6, 6	7 to 10	18, 11, 16, 25	7, 12, 17, 15	0	0	73.7	37.1

Table 9: Per-seed milestones and artifact counts (seeds 3201 to 3204 in order) and condition means of the collaborative artifact fraction and the mean distance travelled per agent. Milestone counts for the paper's ablation episodes are not in the released per-episode table.

Two behavioural differences stand out. The milestone ladder is steeper with the 27B: full culture reaches all sixteen milestones in every seed where the paper's model reaches fourteen, the ablations reach eleven or twelve, and independent search reaches six where the paper's model reaches seven to ten. The 27B is thus the weaker solo agent and gains more from the society, a wider collective gap in the paper's own terms. And the 27B builds far more collaboratively, 0.99 of full-culture artifacts with more than one author against 0.67, and 0.57 to 0.61 in the ablations against 0.22 to 0.26, while travelling about 1.7 times as far per agent. The cross-agent program-fork fraction is not compared, because its definition changed between the studies' code and the release (Section 4).

7.3 The authors' analysis pipeline on our traces

The authors' tools run unchanged on our traces, and their figures for the fifty-agent studies are reproduced here from our episodes. Figure 5 is their study overview for study 1: functional and causally distributed discovery succeed in four of four full-culture episodes and none of four independent-search episodes; field performance is level between the two (0.53 against 0.58); and spontaneous specialisation, the excess normalised mutual information between agents and task roles, is 0.036 with culture and zero without.

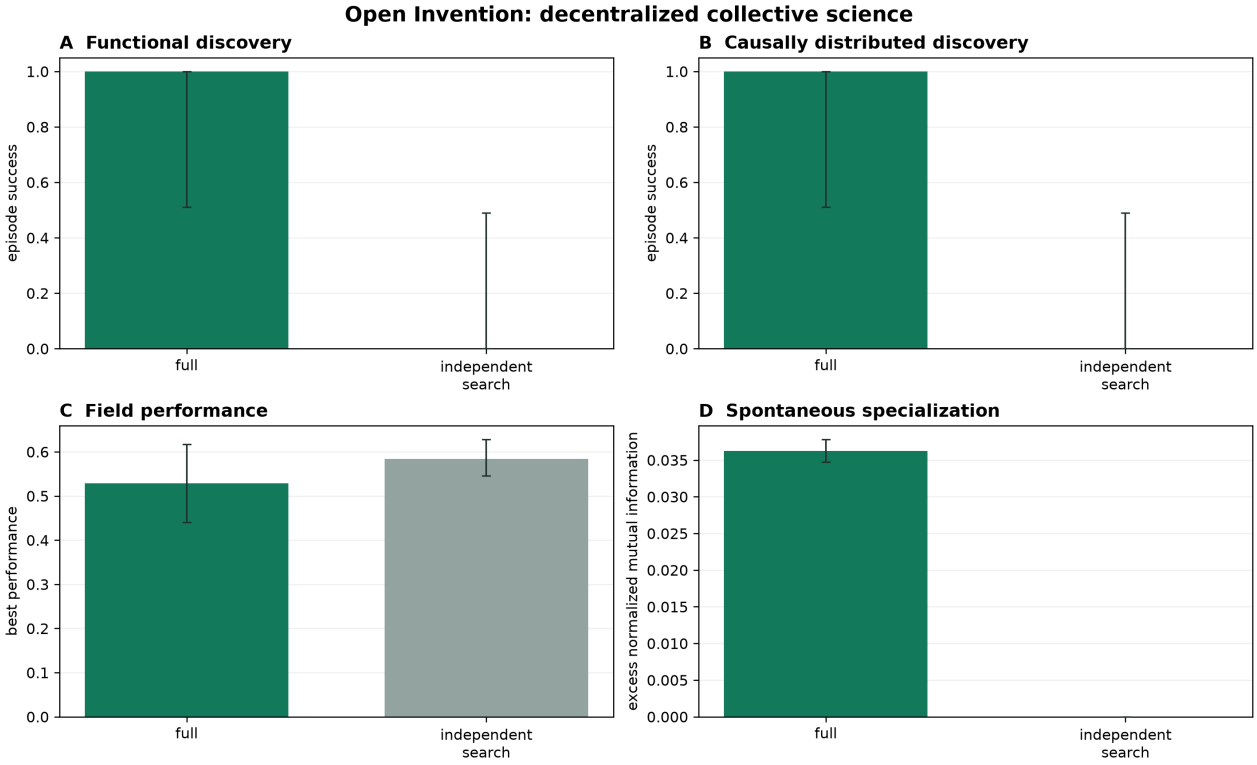


Figure 5: The authors' study overview for our study 1 (full culture and independent search, N = 50). Produced by `analyze-study` on our traces.

The technology-ecology summary (Figure 6) gives 99 % of full-culture artifacts multi-agent ancestry (0 % without culture) and 61 % of program forks cross-agent; independent searchers travel 74 cells per agent against 53 with culture. Frozen-society resilience is about 0.04 in both conditions, so the built technologies do not yet serve the habitat without their agents, as in the paper's N = 50 episodes.

Figure 7 is the authors' lineage figure for the seed that built most. Citation flow between all fifty authors is dense from about tick 300; the 35 artifacts appear between ticks 207 and 780 along the citation chains, 32 of them with cross-agent epistemic lineage; peak performance runs from 0.04 to 0.63 with most above the 0.15 target, and lineage depth and performance are uncorrelated, as the paper argues. The technology dossiers for the same society rank its five best technologies, all mineral, chitin and mycelium resilience matrices built between ticks 305 and 772 by four different agents, peak performance 0.59 to 0.63.

The paper's spatial result reproduces, and Figure 8 shows it on one seed before the statistics (Figures 9 and 10, Table 10). Full-culture societies concentrate in space and around their artifacts; the two ablations spread out like sets of individuals even though they still build; independent searchers wander. Full culture's movement signature is direct and efficient (path directness +0.77, net displacement +0.64, exploration efficiency +0.58 in robust-scaled units), the ablations sit near zero on every axis, and path-length inequality is highest with full culture: a few agents travel, most stay near the work.

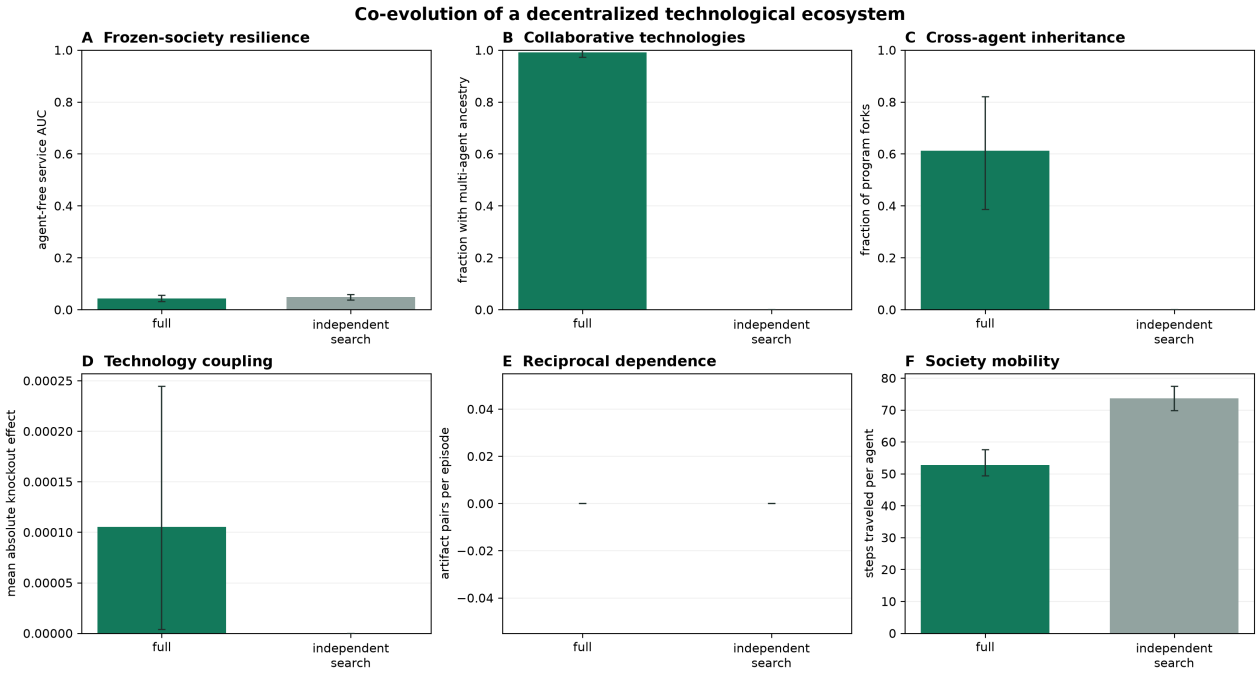


Figure 6: The authors' technology-ecology summary for our study 1: multi-agent ancestry of artifacts, cross-agent program forks, travel per agent and frozen-society resilience.

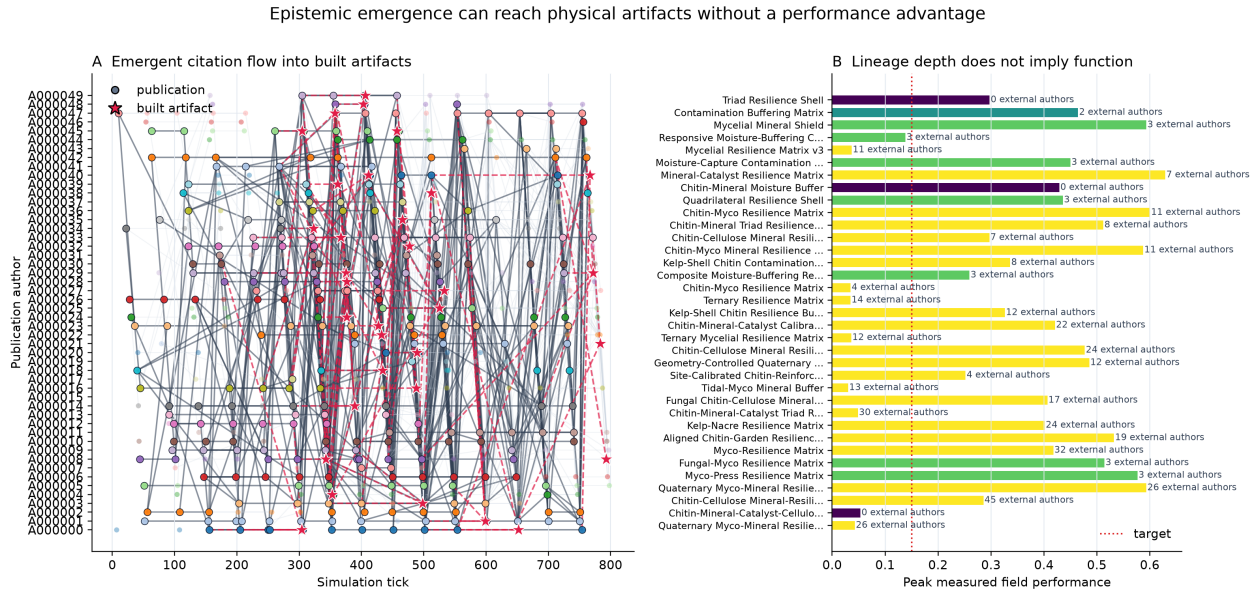


Figure 7: Technology lineage in the seed 3203 full-culture society, which built 35 artifacts. (A) Citation flow between the fifty authors with artifacts (stars) by tick of creation. (B) The 35 artifacts ranked by peak field performance with the number of external authors in each one's ancestry. Produced by `trace-report`.

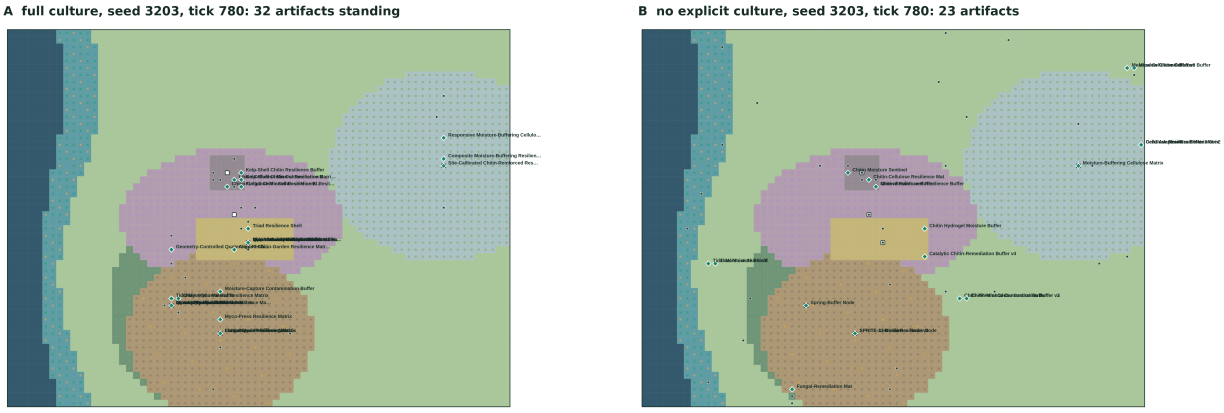


Figure 8: The same world seed under two conditions at tick 780, from the viewer: (A) full culture, 32 artifacts standing; (B) no explicit culture, 23. With culture the society and its artifacts concentrate in a few regions; without it the artifacts are spread across the map and the agents with them.

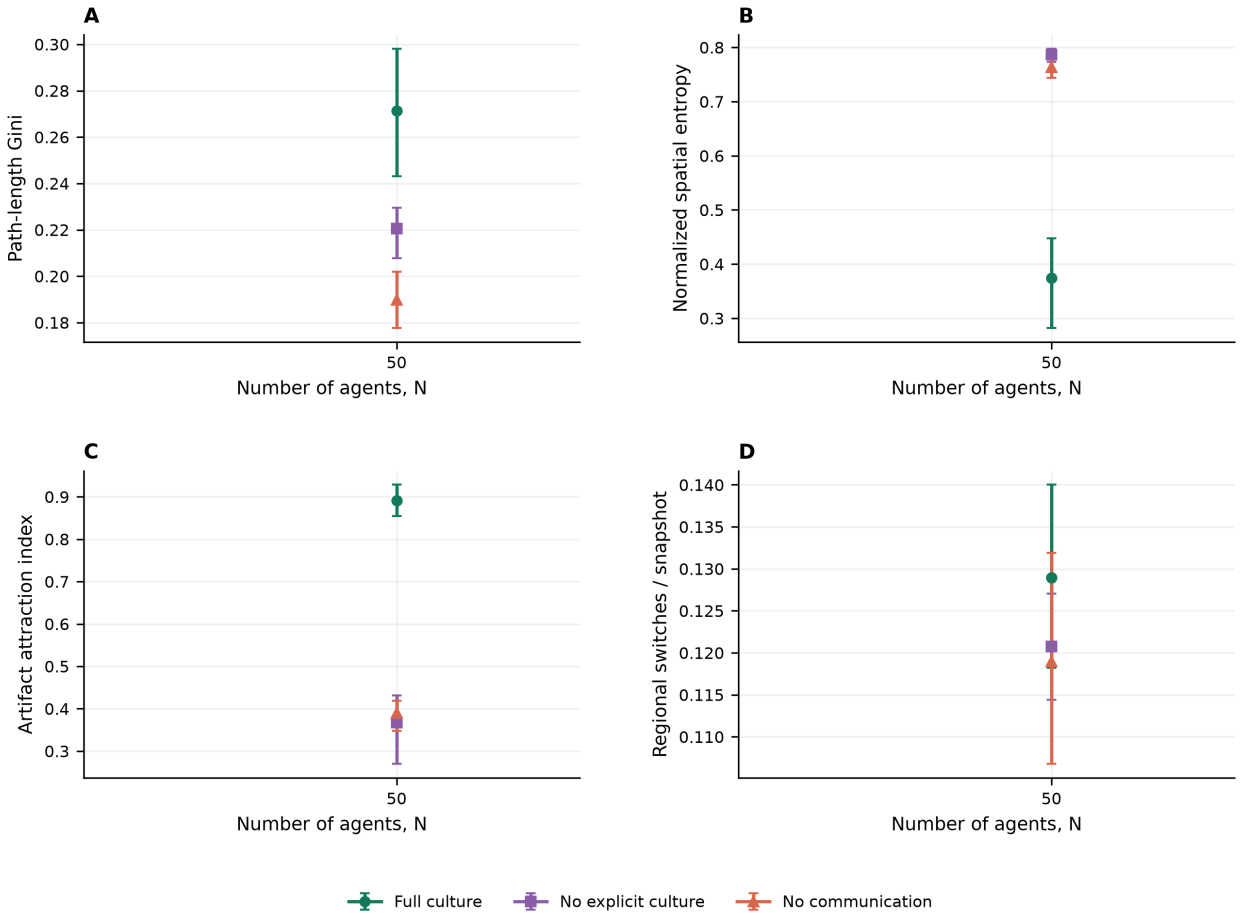


Figure 9: Spatial self-organisation at N = 50 across the four conditions, from the authors' mobility analysis on both studies: spatial entropy, artifact attraction and regional crowding over the episode.

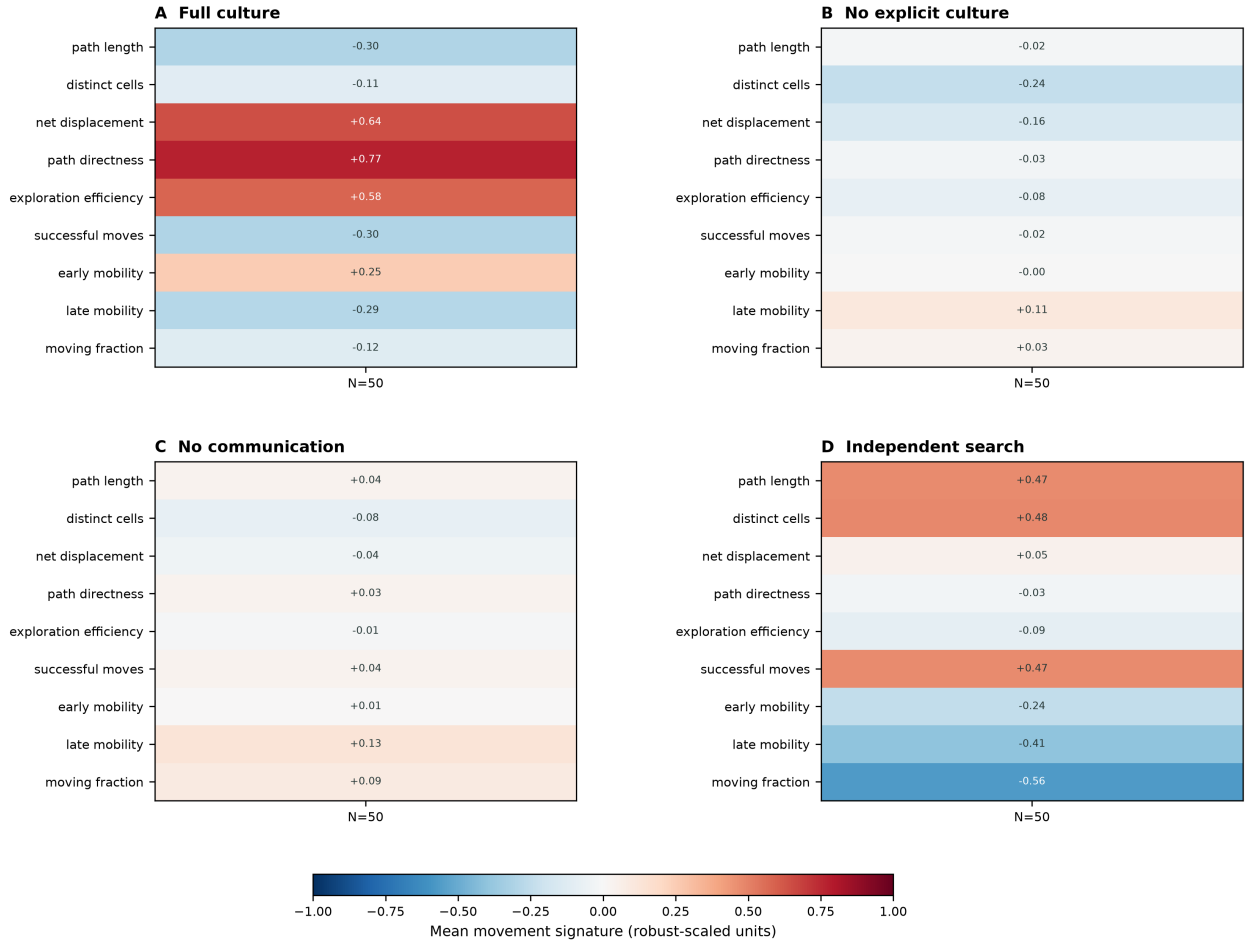


Figure 10: Movement signatures per condition in robust-scaled units, from the same analysis.

condition	spatial entropy (normalised)	artifact attraction index	regional crowding	path-length Gini
full culture	0.37	0.89	0.35	0.27
no communication	0.76	0.39	0.05	0.19
no explicit culture	0.79	0.37	0.04	0.22

Table 10: End-of-episode spatial measures, mean over four seeds, from the authors' mobility analysis on both studies. Independent search has no shared space, so these measures are undefined for it.

7.4 Quantisation and KV-cache precision

The same weights in two 4-bit builds gave validity-pilot means of 0.24 and 0.41. Cell by cell, on the same sixteen conditions, sizes, seeds and prompts, the GPTQ build is higher in 11 cells, equal in 5 and lower in none (Table 11); it builds 8 artifacts in four conditions where the INT4 build builds 2 in one. Both are single samples per cell at temperature 0.7, but an 11-0 split over sixteen paired cells is not sampling noise, and the difference exceeds the one we measured between the 27B and the 235B.

condition	N = 4, seed 3001	N = 4, seed 3002	N = 16, seed 3001	N = 16, seed 3002
full culture	0.19 [0.125]	0.44 [0.25]	0.44 [0.19]	0.44 [0.31]
no communication	0.125 [=]	0.25 [=]	0.50 [=]	0.69 [0.625]
no program forking	0.19 [0.125]	0.31 [=]	0.94 [0.31]	0.69 [0.375]
independent search	0.06 [=]	0.25 [0.06]	0.375 [0.06]	0.625 [0.125]

Table 11: Research score per cell, GPTQ Int4 build with the INT4 w4a16 build in brackets ("=" where equal). GPTQ higher in 11 cells, equal in 5, lower in 0.

The two builds differ in two things at once. The weight recipes are comparable on paper (both 4-bit symmetric, group size 128; the INT4 build leaves the vision encoder, the output head and parts of the linear-attention layers in bf16, the GPTQ build quantises everything but the output head with error compensation), and we cannot rule out that the weights alone account for the split. The one serving difference we know to have changed the arithmetic is the KV cache: the INT4 checkpoint ships a fp8 cache scheme with static per-tensor scales, which vLLM honoured, so every token of its 5k to 15k-token contexts was stored and read back at 8-bit precision, while the GPTQ build's cache was bf16. The symptoms fit precision loss in context rather than in weights: the INT4 build wrote shorter plans (885 against 1,162 output tokens per call) using fewer of the schema's verbs, and its deficit grew in the N = 16 cells, whose prompts are longer. A control run of the INT4 checkpoint with its cache forced to bf16 on the same sixteen cells separates the two causes; it is prepared and listed as work in progress (Section 10). Whatever the split, the practical conclusion stands on the evidence here: on a long-horizon agentic task the KV-cache precision that a checkpoint silently selects is a serving parameter that must be reported next to the weight quantisation, because it moved the same model two rungs on the ladder.

7.5 Throughput and errors

Table 12 gives the serving rates behind Table 6, measured at the server and at the simulator.

Errors are rare and classifiable from the traces. In the N = 50 row, 33 of the 36 errors are runaways at the 4,096-token output cap: well-formed plans whose message fields grow into long narrative late in full-culture episodes, 28 of them in one seed (3204); 2 are gateway timeouts after 900 s caused by KV-cache preemption when two 15k-token requests overflowed one card's cache; 1 occurred in study 2. The paper's own model recorded 16 errors over its 6,415 calls in study 1. In the pilots, the GPTQ build's 29 errors are client timeouts of requests queued inside the server (four streams on a cache that holds 1.5), the INT4 build's single error is a runaway, and the 235B's 11 are 9 schema violations on one action type (an operate step carrying a recipe object) and 2 timeouts. None is a failure of the engine, and the engine's retry policy absorbed all of them without a lost macroturn except the two timeouts of the 235B pilot.

7.6 The marketplace path

Through the Vector marketplace the same model, server and settings answered the simulator with the gateway in between. The five-probe acceptance test passes on both paths. A quick pilot through the gateway (32 calls, no errors) behaved like the direct pilot at the same seed: 329 executed and 9 rejected actions against 320 and 11, research score 0.19 with 3 milestones on both paths, 5.7k input and 1.22k output tokens per call against 5.0k and 1.16k, and no reasoning text leaked through the server-side template. The result so far is that the gateway is behaviourally transparent at pilot scale and that per-request settlement, not the model path, sets its throughput.

build and serving	prompt tokens per call	output tokens per call	decode tokens / s per stream	seconds per call	streams	calls per hour
Gemma 4 E4B, A5000	3.8k	0.33k	about 45	6.5	1	about 550
Qwen3-235B Q2, four-card mesh	3.6k to 6.5k	0.9k to 1.3k	21 to 22 (prompt 305)	32 to 133, mean 71	1	about 51
Qwen3.8-27B INT4, A5000	5.2k	0.89k	about 33	27 (sequential); about 11 aggregate at 4 streams	4	331
Qwen3.8-27B GPTQ, A5000	4.6k to 8.4k	1.1k to 1.3k	about 34	34 (sequential); about 20 aggregate at 4 streams, queued	4	180
Qwen3.8-27B GPTQ, four RTX 3090s, N = 50	15.1k (full), 5.9k to 6.2k (others)	1.65k (full), 1.25k (others)	about 83 per card at batch 2	74 (full), 47 to 52 (others)	8	about 390 (study 1), about 700 (study 2)

Table 12: Throughput. Decode rate is per stream unless marked per card; seconds per call is wall-clock as seen by the simulator, including queueing. Aggregate calls per hour is what sets a study's duration: the fifty-agent full-culture prompts are 15k tokens and dominate study 1.

8 Discussion

8.1 What reproduces

The paper's N = 50 row rests on seed-paired effects, and nine of the twelve reproduce in sign with a 4-bit 27B served on four consumer cards. The ones the paper leans on most are among them: portfolio resilience is positive for every culture condition, the discovery-frontier deficit of full culture and no communication at this population size appears with the same magnitude, and the best-artifact deficit of full culture against the independent envelope is the same size on both sides. The paper's spatial self-organisation result reproduces in full: a full-culture society concentrates around its artifacts, the two ablations spread out like sets of individuals while still building, and independent searchers wander. The authors' analysis pipeline runs unchanged on our traces and produces their figures with the same qualitative content, which is itself a result: the mechanism account survives a change of model family, size, weight precision and serving stack.

The ladder also reproduces the paper's premise that the effects need a model that reaches the build stage. Gemma 4 E4B, the one local profile the authors ship, does not reach it, and its conditions do not separate because there is nothing cultural to lose. The two models that build show the collective contrast; the one that completed the row shows it at the paper's scale.

8.2 What differs

Three differences are not noise and deserve attention in any extension of the study.

Absolute outcomes are higher with the 27B than with the paper's model, in every condition. Best artifact performance, discovery frontier, held-out and portfolio resilience are all above the released values, and in the independent-search control as well, so this is not a cultural effect. Artifact counts, by contrast, are level. The 27B therefore builds about as much and builds better by the engine's physics score. We do not have an explanation that the data can support. Candidates are the longer plans the 27B writes (8 to 10 executed actions per call against the paper's 5.5, so more operations per decision opportunity), the different sampling defaults, and the fact that the two models were served months apart against the same engine revision but not the same code (Section 4). The seed-paired design is what makes the comparison of

effects meaningful despite this shift, and it is why we recommend reading Table 8 rather than Table 9 as the replication result.

The collective gap is wider. The paper's model reaches fourteen milestones with full culture and seven to ten alone; the 27B reaches sixteen with full culture and six alone. A weaker solo agent gains more from the society, which is consistent with the paper's mechanism account and suggests that the size of the cultural effect depends on the model's solo competence, a dimension the paper does not vary. Running the row with two or three models of different solo strength would test this directly.

The 27B collaborates and travels more. Nearly every full-culture artifact has more than one author (0.99 against 0.67), the ablations keep collaborative fractions of 0.57 to 0.61 against 0.22 to 0.26, and agents travel about 1.7 times as far. Part of this is the longer plans; part may be that a model trained more recently on agentic data treats the schema's social verbs as first-class. Either way, the paper's claim that stigmergy through artifacts carries coordination when messages are removed is stronger here, not weaker.

8.3 Serving numerics as a reportable parameter

The second finding was not planned. Two 4-bit builds of the same weights, chosen only because the first was the first one found, differ by 0.17 in mean validity score and split 11-0 cell by cell, a larger gap than the one between the 27B and a 235B. The authors' local-model guidance already asks that the model repository, server, quantisation and context length be recorded; our result adds the KV-cache precision, which neither build's name mentions and which vLLM chose from a field inside one checkpoint's configuration. Had the INT4 build been the only one tried, the conclusion of this paper would have been that a 27B does not play the game. We therefore suggest, for any study that compares models through an inference server, that the trace header or its equivalent record the cache data type and the attention backend next to the weight format, and that comparisons between builds of one model be treated as comparisons between serving configurations until a control isolates the cause. The control for this case, the INT4 checkpoint with its cache forced to bf16 on the same sixteen cells, is prepared (Section 10).

8.4 A cheap replication recipe

The row that the paper ran through a hosted API took about 26 hours on four 24 GB cards and one CPU host, with the authors' code and commands unchanged and two configuration options added. Nothing in the recipe is specific to our fleet: one card per two concurrent episodes, a balancer, a per-cell runner, the guidance grammar backend, thinking off, and the two flags. Section 3 and Appendix C give it in full. At this cost the paper's remaining studies (N = 100, N = 200 and the 3,200-tick crossover) are within reach of a small lab, and so is the design the discussion above calls for, the same row with several models of different solo strength.

9 Threats to validity

Two development seeds in the pilots. The validity pilots rank models on sixteen cells over two seeds. They can rank and screen; they cannot establish the direction of a condition effect, and we draw no such conclusion from them. The 11-0 paired split between the two 27B builds is a within-cell comparison and does not depend on seed count, but its magnitude (0.17 in mean score) is a two-seed estimate.

One sample per cell. Every episode, ours and the paper's, is one sample at temperature 0.7. The paper's N = 50 effects are four-seed means with bootstrap intervals, and ours are the same; the sign agreement in Table 8 is between two four-seed means per effect. Where an effect is within a few hundredths of zero on either side, agreement or disagreement is not informative, which is why the three disagreements are reported with their sizes rather than counted as failures.

The quantisation and KV-cache confound. The two 27B builds differ in weight recipe and in cache precision at once. Section 7.4 states the evidence for the cache as the cause and states that it is not isolated; no claim in this paper depends on which of the two it is.

Code drift within engine revision 9. The paper's traces were produced on six private commits that the public release does not contain, and the hidden state digest of six of seven released traces does not replay on the release (Section 4). Everything we compare against (events, prompts, per-episode metrics) does replay, and our own traces replay bit-exactly on our image. The cross-agent program-fork fraction is the one metric whose definition is known to have changed, and it is excluded from the comparison.

Sampling defaults. The two 27B checkpoints apply top-k 20 and top-p 0.95 under the request temperature; the paper's hosted calls carry no top-k. Both pilots and the row ran with the checkpoint defaults, so the within-27B comparison is unaffected; the comparison with the paper's model carries this difference along with the model difference.

Prompt growth and runaways. Full-culture prompts at N = 50 are about 15k tokens with the 27B against about 13k with the paper's model, because the 27B writes longer messages and records; 33 of its 36 errors in the row are replies that hit the 4,096-token output cap late in full-culture episodes, 28 of them in one seed. The engine's retry handled every one of them without losing a macroturn, as it handled the paper's own 16 errors, but a model that writes more under the same budget plays a slightly different game at the margin.

Timeout drift in one pilot. Fifteen of the sixteen cells of the 235B validity pilot ran with a 180 s client timeout instead of 900 s because a stale rendered configuration was synced over the run folder; two calls were lost to it and retried. The merged summary records the drift. The affected run is a pilot that ranks models, not a result the paper's claims depend on.

Fork fractions and the one unpublished pilot condition. The cross-agent program-fork fraction is not comparable across code versions. The validity pilots' fourth condition is no program forking, which the paper does not publish at N = 50; it is reported only within the ladder.

The marketplace comparison is incomplete. The marketplace arm has a quick pilot and a partial validity pilot. Behavioural equivalence is established at pilot scale; the full comparison is work in progress and no result in Section 7.2 depends on it.

One lab, one fleet. All runs used one package on one fleet. The package is Docker-only and the serving details are recorded per checkpoint, so the recipe is portable, but it has not yet been run by anyone else.

10 Open questions, recommendations and work in progress

10.1 To the authors

1. **Which field entered the state digest between the study commits and the release?** Publishing the diff of the relevant module, or the private commits, would let the released traces verify as documented; failing that, a note in the data card that exact replay is expected to fail the hidden digest late while snapshots, events and metrics reproduce would save every replicator the diagnosis in Section 4.
2. **Serving parameters in the method.** The local-model guidance asks for repository, server, quantisation and context length. We suggest adding the KV-cache data type and the attention backend, and that the engine record the server's reported model name and version in the trace header, as our run manifests do.
3. **Solo competence as a factor.** The wider collective gap with a weaker solo agent (Section 8) suggests that the size of the cultural effect depends on the model's unaided ability. The paper's design varies the mechanisms and the population; varying the model is the obvious next axis, and the local-model profile already in the repository makes it cheap.
4. **The output budget.** Models that write more hit the 4,096-token cap late in full-culture episodes. A schema-level bound on message and record lengths, or a documented per-model budget, would make the cap a less model-dependent part of the task.

10.2 To the community

The recipe of Section 8 reproduces the paper's main comparison on four consumer cards in about a day, with the authors' code and commands unchanged. Replications on other fleets, with other open-weight models, and of the studies this paper did not run would turn a single replication into a body of evidence about which of the paper's findings are robust to the model. The two configuration options are in the authors' repository as a pull request and in our package behind flags; the acceptance test for structured-output pass-through applies to any OpenAI-compatible gateway.

10.3 Work in progress

- **Control arm for the KV-cache finding.** The INT4 checkpoint with its cache forced to bf16, on the same sixteen validity cells. If its score rises to the GPTQ level the cache was the cause; if not, the weights were. About half a day on one 24 GB card; the configuration is in the package registry.
- **The paper's model on our infrastructure.** An N = 50 row with gpt-5.6-luna through our runner, so that the comparison with the 27B is same-seed and same-package rather than same-seed and same-engine-revision.
- **The marketplace comparison.** The validity pilot through the Vector marketplace, paired with the direct run, once the gateway's per-request settlement is replaced by a per-run allocation.
- **N = 100.** The paper's next population size, where its conditions overtake the independent envelope on the discovery frontier; about two days on the same pool.

11 Conclusion

Using the SwarmWorld authors' released code and commands unchanged, with two opt-in configuration options for local servers, we reproduced the paper's N = 50 comparison on four 24 GB consumer cards with an open-weight 27B model at 4-bit weight precision. Nine of the twelve seed-paired effects agree in sign with the paper's, the clearest of them in magnitude as well; the spatial self-organisation result reproduces in full; and the authors' analysis pipeline, applied to our traces, draws the same picture. The 27B is the weaker solo agent and the stronger society member, which widens the paper's collective gap rather than closing it.

Two findings came out of the way, not the destination. The same weights in two 4-bit builds differ by two rungs on the model ladder, with the precision of the KV cache that one checkpoint silently selects as the leading suspect; and the paper's released traces do not replay their hidden state digest on the released engine although everything observable in them does, which points to code drift within one declared engine revision. Both are reported upstream with evidence. Both argue for the same thing: a replication of agentic simulation results needs the serving configuration and the exact engine recorded with the same care as the seeds, and the package that produced this paper is an attempt to show what that looks like.

Acknowledgements

We thank Subhadeep Pal, Fiona Y. Wang and Markus J. Buehler for releasing the SwarmWorld engine, the analysis tools and the complete study traces that made a seed-paired replication possible, and for the documentation that let the authors' own commands be run unchanged. We thank the HAL8 marketplace team for integrating a locally served model as a new type of supplier on the Vector marketplace, which gave this study its second serving path.

AI assistance was used throughout this work and is acknowledged as part of its method: Claude (Anthropic), through Claude Code, assisted with planning the replication, writing the package and its scripts, diagnosing the engine and serving behaviour described in Sections 4 and 5, analysing the traces and drafting this text. All runs were launched by the author on the lab fleet, every number in the paper is computed from the recorded traces by the authors' tools or by the package scripts, and the author reviewed and takes responsibility for the whole.

Data and code availability

The replication package (Docker Compose files, the vendored engine with the two patches behind flags and the pristine files beside them, the runner, balancer and viewer services, the model registry, the acceptance test for structured-output gateways, and the analysis scripts) is available from the author and will be published alongside the trace mirror. The traces of all sixteen episodes of the N = 50 row, the four validity pilots and the quick pilots, with their run manifests and the outputs of the authors' analysis pipeline, are mirrored on Hugging Face as `apexfusion/swarmworld-replication-data` (HAL8 AI Local Lab 2026) (https://huggingface.co/datasets/apexfusion/swarmworld-replication-data) in the layout of the original dataset (a file manifest with hashes, an episode table and a study table), under the Apache 2.0 licence of the original. The two engine changes are in the authors' repository as pull request #2 and the replay finding as issue #1.

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Appendix A: model registry

The registry records, per served alias, what the traces cannot: the checkpoint, its quantisation, the serving stack and the measured side. The package joins the measured columns automatically from the run manifests and study summaries.

alias	checkpoint	quantisation	loaded weights	server, kernels	context	KV cache	notes
gpt-5.6-luna	hosted (OpenAI)	none	n/a	OpenAI Responses API, reasoning effort low, temperature 0.7	60,000-character prompt budget, 4,096 output tokens	n/a	the paper's model; released traces used as the reference

alias	checkpoint	quantisation	loaded weights	server, kernels	context	KV cache	notes
swarm-gemma-4-e4b	google/gemma-4-E4B-it	bf16	15.2 GiB	vLLM 0.29.0; guidance grammar with whitespace disabled; max-num-seqs 8 to 16; GPU memory utilisation 0.90	32,768	bf16, 5.57 GiB = 269,718 tokens	the authors' local profile; floor of the ladder
qwen3-235b-q3kxl	unsloth/Qwen3-235B-A22B-GGUF, file UD-Q2_K_XL (alias from an earlier Q3 trial)	GGUF UD-Q2_K_XL, 2-bit dynamic	about 88 GB over four cards	llama.cpp llama-server, RPC pipeline over four nodes, tensor split 17,17,20,20, layer split, flash attention	32,768	q4_0 keys and values; one slot (two slots: 7.35 tok/s aggregate)	prompt 305 tok/s, decode 21 to 22 tok/s; model load about 15 min per restart
swarm-qwen38-27b-int4	RedHatAI/Qwen3.8-27B-INT4	compressed-tensors w4a16, 4-bit symmetric, group 128, Marlin; vision encoder, output head and parts of the linear-attention layers in bf16	16.84 GiB text-only (17.71 GiB with the vision encoder)	vLLM 0.29.0, FlashInfer; GPU memory utilisation 0.95, max-num-seqs 8, multimodal limits set to zero	24,576 (32,768 did not fit)	fp8 e4m3 from the checkpoint's KV scheme (static per-tensor scales), 2.85 GiB = 65,908 tokens, 2.68 concurrent 24k requests	generation config top-k 20, top-p 0.95 under the request temperature
swarm-qwen38-27b-gptq4	Max73333/Qwen3.8-27B-GPTQ-Int4-V100	GPTQ Int4 (GPTQModel 7.3.2), 4-bit symmetric, group 128, desc_act off, damp 0.05, output head not quantised; Marlin	16.83 GiB	vLLM 0.29.0, FlashAttention 2; same memory settings; quantisation auto-detected	24,576	bf16, 2.82 GiB = 37,510 tokens, 1.53 concurrent 24k requests	the model of the N = 50 row; same sampling defaults as the INT4 build
swarm-qwen38-27b-int4-bf16kv	RedHatAI/Qwen3.8-27B-INT4	as swarm-qwen38-27b-int4 with the cache forced to bf16	16.84 GiB	vLLM 0.29.0	24,576	bf16 (expected about 37.5k tokens)	control arm, not yet run

alias	checkpoint	quantisation	loaded weights	server, kernels	context	KV cache	notes
`swarm-qwen38-27b-awq`	`barrydeen/Qwen3.8-27B-AWQ-4bit`	AWQ 4-bit (compressed-tensors metadata)	does not load	vLLM 0.29.0	24,576 requested	n/a	22 GiB of bf16 tensors allocated before the output head: quantisation config does not match the hybrid module names

Table 13: Model registry. Calibration data are not documented for either 27B build.

Appendix B: per-cell results

Every episode behind Sections 7.1 and 7.2, from the study summaries written by the authors' `research-study` command. Columns: model calls, errors, actions executed and rejected, research score, milestones reached (ms.), artifacts built (art.), best built artifact's field performance (best; 0 where nothing was built), input and output tokens per model call.

Gemma 4 E4B, validity pilot

condition	N	seed	calls	err.	exec.	rej.	score	ms.	art.	best	in/call	out/call
full	4	3001	32	0	37	6	0.125	2	0	0.00	3893	302
full	4	3002	32	0	33	2	0.062	1	0	0.00	3884	270
full	16	3001	128	0	153	22	0.250	4	0	0.00	3987	312
full	16	3002	128	0	153	26	0.125	2	0	0.00	4436	315
independent-search	4	3001	32	0	43	10	0.062	1	0	0.00	3710	352
independent-search	4	3002	32	0	48	15	0.062	1	0	0.00	3718	372
independent-search	16	3001	128	0	157	33	0.062	1	0	0.00	3693	325
independent-search	16	3002	128	0	188	36	0.125	2	0	0.00	3767	343
no-communication	4	3001	32	0	39	7	0.125	2	0	0.00	3632	310
no-communication	4	3002	32	0	39	7	0.125	2	0	0.00	3712	312
no-communication	16	3001	128	0	163	38	0.250	4	0	0.00	3736	332
no-communication	16	3002	128	0	170	33	0.125	2	0	0.00	3811	339
no-explicit-culture	4	3001	32	0	41	4	0.125	2	0	0.00	3553	316
no-explicit-culture	4	3002	32	0	41	13	0.125	2	0	0.00	3701	372
no-explicit-culture	16	3001	128	0	178	37	0.250	4	0	0.00	3580	325
no-explicit-culture	16	3002	128	0	199	30	0.250	4	0	0.00	3720	362

Qwen3-235B-A22B UD-Q2_K_XL, validity pilot

condition	N	seed	calls	err.	exec.	rej.	score	ms.	art.	best	in/call	out/call
full	4	3001	32	0	187	38	0.438	7	5	0.19	5517	1227
full	4	3002	32	3	155	31	0.312	5	0	0.00	4727	1025
full	16	3001	129	1	814	136	0.438	7	2	0.07	6114	1145
full	16	3002	128	0	876	152	0.500	8	1	0.07	6535	1322
independent-search	4	3001	32	0	223	30	0.188	3	1	0.04	3741	1239
independent-search	4	3002	32	2	151	25	0.188	3	1	0.03	3581	975
independent-search	16	3001	128	1	754	77	0.375	6	3	0.31	3799	1056
independent-search	16	3002	128	0	779	119	0.375	6	8	0.20	4085	1186
no-communication	4	3001	32	0	144	16	0.188	3	1	0.03	3855	872
no-communication	4	3002	32	0	189	40	0.125	2	0	0.00	3698	1174
no-communication	16	3001	128	2	692	94	0.438	7	8	0.06	3930	1082
no-communication	16	3002	128	0	724	165	0.438	7	8	0.05	3972	1118
no-program-forking	4	3001	32	0	193	37	0.250	4	1	0.02	4869	1107
no-program-forking	4	3002	32	0	134	36	0.438	7	2	0.20	4942	950
no-program-forking	16	3001	128	0	677	148	0.438	7	2	0.06	5989	1086
no-program-forking	16	3002	129	2	720	221	0.625	10	3	0.38	6296	1211

Qwen3.8-27B INT4 w4a16 (fp8 KV), validity pilot

condition	N	seed	calls	err.	exec.	rej.	score	ms.	art.	best	in/call	out/call
full	4	3001	32	0	261	10	0.125	2	0	0.00	4620	931
full	4	3002	32	0	189	11	0.250	4	0	0.00	4842	829
full	16	3001	128	0	1006	23	0.188	3	0	0.00	4905	899
full	16	3002	128	0	748	19	0.312	5	0	0.00	6348	872
independent-search	4	3001	32	0	269	0	0.062	1	0	0.00	4296	948
independent-search	4	3002	32	0	181	12	0.062	1	0	0.00	4204	724
independent-search	16	3001	128	0	1068	9	0.062	1	0	0.00	4387	916
independent-search	16	3002	128	0	807	37	0.125	2	0	0.00	4499	812
no-communication	4	3001	32	0	246	1	0.125	2	0	0.00	4264	888
no-communication	4	3002	32	0	207	10	0.250	4	0	0.00	4251	761
no-communication	16	3001	128	1	1039	34	0.500	8	1	0.26	4262	884
no-communication	16	3002	128	0	850	33	0.625	10	1	0.23	4520	829
no-program-forking	4	3001	32	0	252	1	0.125	2	0	0.00	4621	912
no-program-forking	4	3002	32	0	208	10	0.312	5	0	0.00	5008	854
no-program-forking	16	3001	128	0	999	25	0.312	5	0	0.00	7309	1048

condition	N	seed	calls	err.	exec.	rej.	score	ms.	art.	best	in/call	out/call
no-program-forking	16	3002	128	0	851	47	0.375	6	0	0.00	6397	907

Qwen3.8-27B GPTQ Int4 (bf16 KV), validity pilot

condition	N	seed	calls	err.	exec.	rej.	score	ms.	art.	best	in/call	out/call
full	4	3001	32	0	329	8	0.188	3	0	0.00	5771	1312
full	4	3002	32	0	268	42	0.438	7	1	0.03	5627	1135
full	16	3001	134	6	1251	85	0.438	7	0	0.00	5879	1127
full	16	3002	133	5	1041	125	0.438	7	0	0.00	8419	1304
independent-search	4	3001	32	0	313	9	0.062	1	0	0.00	4576	1165
independent-search	4	3002	32	0	263	35	0.250	4	0	0.00	4895	1142
independent-search	16	3001	130	2	1254	39	0.375	6	1	0.37	4628	1143
independent-search	16	3002	129	1	1152	57	0.625	10	1	0.29	4886	1179
no-communication	4	3001	32	0	324	2	0.125	2	0	0.00	4613	1181
no-communication	4	3002	32	0	269	17	0.250	4	0	0.00	4820	1096
no-communication	16	3001	128	0	1228	56	0.500	8	1	0.31	4649	1145
no-communication	16	3002	130	2	1086	64	0.688	11	2	0.37	4864	1128
no-program-forking	4	3001	32	0	315	11	0.188	3	0	0.00	5580	1246
no-program-forking	4	3002	33	1	285	29	0.312	5	0	0.00	5225	1164
no-program-forking	16	3001	136	8	1186	66	0.938	15	1	0.30	7758	1265
no-program-forking	16	3002	132	4	1108	115	0.688	11	1	0.04	7841	1326

Qwen3.8-27B GPTQ Int4, N = 50 study 1

condition	N	seed	calls	err.	exec.	rej.	score	ms.	art.	best	in/call	out/call
full	50	3201	800	1	6043	1109	1.000	16	12	0.60	15132	1700
full	50	3202	802	3	6437	983	1.000	16	27	0.47	14772	1670
full	50	3203	800	3	5887	633	1.000	16	35	0.63	14984	1615
full	50	3204	800	28	5517	930	1.000	16	12	0.41	15216	1611
independent-search	50	3201	800	0	7779	247	0.375	6	18	0.65	5822	1255
independent-search	50	3202	800	0	7533	402	0.375	6	11	0.53	5841	1242
independent-search	50	3203	800	0	7479	311	0.375	6	16	0.56	5905	1247
independent-search	50	3204	800	0	7457	216	0.375	6	25	0.60	5896	1247

Qwen3.8-27B GPTQ Int4, N = 50 study 2

condition	N	seed	calls	err.	exec.	rej.	score	ms.	art.	best	in/call	out/call
no-communication	50	3201	800	0	7055	391	0.750	12	12	0.52	6176	1277
no-communication	50	3202	800	0	6778	448	0.750	12	18	0.86	6382	1267
no-communication	50	3203	800	0	6809	393	0.688	11	15	0.87	6369	1249
no-communication	50	3204	800	0	6620	371	0.750	12	28	0.48	6562	1267
no-explicit-culture	50	3201	800	1	6898	315	0.750	12	13	0.87	6046	1241
no-explicit-culture	50	3202	800	0	6506	374	0.750	12	17	0.56	6254	1256
no-explicit-culture	50	3203	800	0	6541	423	0.750	12	23	0.71	6255	1255
no-explicit-culture	50	3204	800	0	6467	377	0.750	12	15	0.46	6082	1244

N = 50 row against the released episodes, per seed

Ours | paper for each endpoint; the seed-paired effect is the condition's value minus independent search on the same seed.

condition	seed	frontier AUC	held-out AUC	best perf.	portfolio res.
full	3201	0.301 | 0.254	0.033 | 0.030	0.605 | 0.365	0.200 | 0.179
full	3202	0.293 | 0.206	0.043 | 0.025	0.469 | 0.337	0.203 | 0.083
full	3203	0.415 | 0.306	0.064 | 0.033	0.629 | 0.472	0.263 | 0.138
full	3204	0.247 | 0.165	0.030 | 0.031	0.411 | 0.357	0.204 | 0.094
no-communication	3201	0.319 | 0.159	0.059 | 0.024	0.518 | 0.368	0.219 | 0.141
no-communication	3202	0.331 | 0.308	0.074 | 0.030	0.861 | 0.508	0.307 | 0.186
no-communication	3203	0.352 | 0.268	0.045 | 0.031	0.868 | 0.455	0.289 | 0.162
no-communication	3204	0.318 | 0.238	0.061 | 0.028	0.483 | 0.325	0.225 | 0.110
no-explicit-culture	3201	0.264 | 0.303	0.059 | 0.027	0.869 | 0.423	0.251 | 0.195
no-explicit-culture	3202	0.330 | 0.301	0.047 | 0.029	0.562 | 0.454	0.232 | 0.176
no-explicit-culture	3203	0.345 | 0.318	0.080 | 0.035	0.711 | 0.431	0.252 | 0.203
no-explicit-culture	3204	0.271 | 0.256	0.057 | 0.034	0.463 | 0.510	0.211 | 0.208
independent-search	3201	0.375 | 0.218	0.055 | 0.013	0.649 | 0.408	0.109 | 0.079
independent-search	3202	0.239 | 0.300	0.034 | 0.015	0.529 | 0.369	0.119 | 0.065
independent-search	3203	0.361 | 0.303	0.038 | 0.025	0.563 | 0.481	0.150 | 0.082
independent-search	3204	0.410 | 0.280	0.062 | 0.022	0.596 | 0.540	0.189 | 0.121

Appendix C: commands and package versions

Every run is launched from the package root with Docker Compose; the simulator container runs on the CPU host and reaches the model servers over the LAN. The commands below are the ones used for the reported runs, with the model alias and server address as environment variables. The authors' `research-study` arguments are unchanged inside the scripts.

```
# Step 0: engine verification (tests, self-replay, dataset hashes, trace replay)
docker compose run --rm sim /work/scripts/step0_verify.sh

# a model server, one per card; identical image, revision and flags everywhere
docker compose --profile vllm up -d vllm        # vLLM 0.29.0; model, flags from .env
docker compose --profile lb up -d lb            # nginx balancer over VLLM_UPSTREAMS

# quick pilot (full culture, N = 4, seed 3001, 400 ticks)
CFG=/work/configs-local/vllm-qwen36-27b.yaml LLM_MODEL=<alias> \
  docker compose run --rm sim /work/scripts/step1a_pilot.sh

# validity pilot, cells in parallel
```

```
docker compose run -d -e CFG=... -e LLM_MODEL=<alias> -e RUN_TAG=<tag> \
  -e STUDIES=validity -e PARALLEL=4 sim /work/scripts/step1b_parallel.sh

# the N = 50 row, one study per launch, eight streams over four cards
docker compose run -d -e CFG=... -e LLM_MODEL=<alias> \
  -e LLM_BASE_URL=http://lb:8080/v1 -e STUDIES=full_independent \
  -e PARALLEL=8 -e RUN_TAG=<tag> sim /work/scripts/step1b_parallel.sh
docker compose run -d ... -e STUDIES=culture_ablations ...

# after the row: the authors' analysis on our traces
docker compose run --rm sim biofoundry analyze-study <study>/study-summary.json \
  --output-dir <study>/analysis
docker compose run --rm sim python /work/scripts/compare_to_paper.py <study1> <study2> \
  --released /work/data/released/metadata/episodes.csv --out <out>.csv
docker compose run --rm sim biofoundry trace-report <trace> --output <json> \
  --figures-dir <dir>
docker compose run --rm sim biofoundry technology-dossiers <trace> \
  --output-dir <dir> --top-k 5
docker compose run --rm sim biofoundry analyze-mobility <study1 summary> \
  <study2 summary> --output-dir <dir>
docker compose run --rm sim biofoundry playback <trace> --host 0.0.0.0 --port 8765
```

run	package version	engine	vLLM	config hash (12)
Gemma 4 E4B validity	2026.09.09-9 (before run manifests)	revision 9, commit 6af7ae9	0.29.0	—
Qwen3-235B validity	2026.09.09-42	revision 9	llama-server (build not recorded)	5f9d4e22a7eb
Qwen3.8-27B INT4 validity	2026.09.09-21	revision 9	0.29.0	af11a0c2d609
Qwen3.8-27B GPTQ validity	2026.09.09-33	revision 9	0.29.0	3503dd35d17a
Qwen3.8-27B GPTQ N = 50, study 1	2026.09.09-44	revision 9	0.29.0 × 4	8978129668be
Qwen3.8-27B GPTQ N = 50, study 2	2026.09.09-48	revision 9	0.29.0 × 4	8978129668be

Table 21: Package and server versions per run, from the run manifests. The package version is the date-serial of the replication package; the engine is the vendored release in every case.